

Clausen constants at roots of unity: the fundamental basis by Dirichlet conductor, and a weight parity swap

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Store `clausen-roots-of-unity.wl`. Companions: `polylog-vertical-line`, `cleo-arctan-sin-chi2`.

Abstract

The imaginary parts $\text{Cl}_2(2\pi a/n) = \text{Im Li}_2(e^{2\pi i a/n})$ that appear when one evaluates the dilogarithm at algebraic points (as in Reshetnikov’s Math.StackExchange questions 1430169 and 1410276) are organized completely by the theory of odd Dirichlet characters. We record the explicit picture and verify it numerically: the $\mathbb{Q}$ -span of $\{\text{Cl}_2(2\pi a/n)\}$ has dimension $\varphi(n)/2$, and the genuinely *new* constants entering at exact level n are indexed by the *primitive* odd characters mod n . By conductor the fundamental constants are $V = \text{Cl}_2(\pi/3)$ (Giesecking, conductor 3), $G = \text{Cl}_2(\pi/2)$ (Catalan, conductor 4), $W_8 = \text{Cl}_2(\pi/4)$ (conductor 8), and $W_{24} = \text{Cl}_2(\pi/12)$ (conductor 24); conductor 12 contributes nothing. This settles the open question in MSE answer a/5113303: level 24 needs exactly *one* new constant W_{24} beyond $\{G, V, W_8\}$. We give the reduction tables, the L -value/trigamma forms, a new closed form $\text{Li}_2(e^{i\pi/12}) = \frac{73\pi^2}{576} + iW_{24}$, and a clean *weight parity swap*: at weight 3 the imaginary part $\text{Im Li}_3(e^{2\pi i a/n})$ is *elementary* (a rational multiple of π^3), so the deep constants move to the real part—the reverse of weight 2.

1 Context: dilogarithms at algebraic points

MSE 1430169 asks for a closed form of $\text{Li}_2(\frac{1}{2} + i(1 - \frac{\sqrt{3}}{2}))$. The standard route (Sangchul Lee, a/1430262) uses $\text{Li}_2(z) = -\text{Li}_2(z/(z-1)) - \frac{1}{2}\ln^2(1-z)$ to send the point to the unit circle: here $z/(z-1) = e^{7\pi i/6}$, and $\text{Li}_2(e^{i\theta}) = (\frac{\pi^2}{6} - \frac{\pi\theta}{2} + \frac{\theta^2}{4}) + i\text{Cl}_2(\theta)$ for $\theta \in [0, 2\pi)$, so everything reduces to the Clausen value $\text{Cl}_2(7\pi/6)$. The closed form’s nontrivial content is therefore the reduction of a single Cl_2 at a rational angle. A later answer (a/5113303) expresses the whole level-12 family through two constants—the Catalan constant G and the Giesecking constant V —and asks whether level 24 needs something new. It does; the organizing principle is classical, and we make it explicit.

2 The fundamental constants, by conductor

For $n > 2$ the values $\text{Cl}_2(2\pi a/n)$, $1 \leq a < n$, are linear combinations of $L(2, \chi)$ over the *odd* Dirichlet characters χ mod n (those with $\chi(-1) = -1$), because Cl_2 is an odd function of the angle. Hence:

Proposition 1. $\dim_{\mathbb{Q}} \text{span}\{\text{Cl}_2(2\pi a/n) : 1 \leq a < n\} = \varphi(n)/2$, the number of odd characters mod n . The constants genuinely new at exact level n (not already present at any proper divisor) are

indexed by the primitive odd characters mod n ; their count is $0, 0, 1, 0, 2, 1, 2, 0, 4, 0, 5, 0, 1, 2, 7, \dots$ for $n = 3, 4, 5, \dots$.

The first few fundamental constants, one per primitive odd character, are

$$\underbrace{V = \text{Cl}_2(\frac{\pi}{3})}_{\text{conductor 3}}, \quad \underbrace{G = \text{Cl}_2(\frac{\pi}{2})}_{\text{conductor 4}}, \quad \underbrace{W_8 = \text{Cl}_2(\frac{\pi}{4})}_{\text{conductor 8}}, \quad \underbrace{W_{24} = \text{Cl}_2(\frac{\pi}{12})}_{\text{conductor 24}}.$$

Conductor 12 has *no* primitive odd character (every odd character mod 12 factors through 3 or 4), which is exactly why the level-12 family of $a/5113303$ collapses onto $\{G, V\}$. Conductors 6 and 10 likewise add nothing. The pentagon level 5 is the first to add *two* constants at once, $\text{Cl}_2(2\pi/5)$ and $\text{Cl}_2(4\pi/5)$ (two conjugate primitive odd characters mod 5).

Their canonical L -value (trigamma) forms are

$$G = \frac{\psi^{(1)}(\frac{1}{4}) - \pi^2}{8}, \quad \sqrt{3}V = \frac{3\psi^{(1)}(\frac{1}{3}) - 2\pi^2}{6} \quad \left(\text{i.e. } \psi^{(1)}(\frac{1}{3}) = \frac{2\pi^2}{3} + 2\sqrt{3}V\right). \quad (1)$$

3 Reduction tables and the settled level-24 question

Every $\text{Cl}_2(2\pi a/n)$ at $n \in \{8, 12, 24\}$ reduces to the basis above (PSLQ, ≥ 110 digits, Champernowne canary; all confirmed in Wolfram):

$$\begin{aligned} \text{level 12 : } & \text{Cl}_2(\frac{\pi}{6}) = \frac{2}{3}G + \frac{1}{4}V, \quad \text{Cl}_2(\frac{5\pi}{6}) = \frac{2}{3}G - \frac{1}{4}V, \quad \text{Cl}_2(\frac{2\pi}{3}) = \frac{2}{3}V; \\ \text{level 8 : } & \text{Cl}_2(\frac{3\pi}{4}) = W_8 - \frac{1}{2}G; \\ \text{level 24 : } & \text{Cl}_2(\frac{5\pi}{12}) = \frac{16W_8 + 24W_{24} - 4G - 3V}{24}, \quad \text{Cl}_2(\frac{7\pi}{12}) = \frac{4W_8 + 6W_{24} - 3G}{6}, \\ & \text{Cl}_2(\frac{11\pi}{12}) = \frac{24W_{24} - 8G - 3V}{24}. \end{aligned}$$

The four level-24 values $\text{Cl}_2(k\pi/12)$, $k \in \{1, 5, 7, 11\}$, span a 4-dimensional space $\{G, V, W_8, W_{24}\}$ ($= \varphi(24)/2$), and W_{24} is provably outside $\text{span}\{G, V, W_8\}$.

Remark 2. This settles $a/5113303$: the dilogarithms of 24th roots of unity require exactly one constant beyond the level-12 pair $\{G, V\}$ and the level-8 constant W_8 , namely $W_{24} = \text{Cl}_2(\pi/12)$.

4 A new closed form

Since $\text{Cl}_2(\pi/12) = W_{24}$ is itself fundamental, the cleanest carrier is the dilogarithm at the primitive 24th root $e^{i\pi/12}$:

$$\text{Li}_2(e^{i\pi/12}) = \frac{73\pi^2}{576} + iW_{24}, \quad (2)$$

the real part being $\frac{\pi^2}{6} - \frac{\pi}{2} \cdot \frac{\pi}{12} + \frac{1}{4}(\frac{\pi}{12})^2$. Pulling this back through $z = 1/(1 - e^{i\theta})$ gives Reshetnikov-style closed forms at the corresponding $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$ points; the cleanest stays the root-of-unity form (2).

5 A weight parity swap

The same character parity that makes Cl_2 (weight 2, odd characters) carry deep constants reverses one weight up. For odd χ , $L(s, \chi)$ is a rational multiple of π^s exactly when s is *odd*; since $\text{Im Li}_3(e^{i\theta}) = \sum \sin(k\theta)/k^3$ pairs with odd characters at $s = 3$,

$$\text{Im Li}_3(e^{2\pi ia/n}) \in \mathbb{Q} \cdot \pi^3 \quad \text{for all } a, n, \quad (3)$$

e.g. $\operatorname{Im} \operatorname{Li}_3(e^{2\pi i/3}) = \frac{2\pi^3}{81}$, $\operatorname{Im} \operatorname{Li}_3(i) = \frac{\pi^3}{32}$, $\operatorname{Im} \operatorname{Li}_3(e^{i\pi/3}) = \frac{5\pi^3}{162}$, $\operatorname{Im} \operatorname{Li}_3(e^{2\pi i/5}) = \frac{4\pi^3}{125}$. Thus at weight 3 the *imaginary* part at roots of unity is elementary, and the deep constants ($\zeta(3)$ and the even-character $L(3, \chi)$) live in the *real* part—the exact reverse of weight 2. (On the line $\operatorname{Re} z = 1$ the picture is different again: there $\operatorname{Im} \operatorname{Li}_3(1+i)$ is the central-binomial Gaussian tower constant, because $1+i$ is not a root of unity; see `polylog-vertical-line`.)

6 Verification and reproducibility

Proposition 1’s reductions, the trigamma forms (1), the new closed form (2), the original MSE value, and the weight-3 rationals (3) are verified in `mpmath` (PSLQ, 90–110 digits, Champernowne canary) and Wolfram, residuals 10^{-63} to 10^{-73} . (Two level-24 reductions were sign-corrected against Wolfram before storing.) Closed forms are in `src/PolyLog/identities/clausen-roots-of-unity.wl`; scripts under `src/PolyLog/tools/scratch/clausen-roots/`.